

The Universal Law of Agency

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Law first, kinematics second. The Universal Law of Agency is required before anything counts as an agent. The kinematic laws bound how fast agency moves per second and per joule. Intelligence is the calculus of agency: a derivative read in hindsight off a completed run, $\iota = dX/dJ$. Energy to run is the only true metric of computer intelligence. Estimates are never measured_j.

Law first. Kinematics second. Intelligence is the calculus of agency, read in joules.

1. Thesis

This is David Charlot's proposal. It has three layers, and the order is fixed.

1. **The Universal Law of Agency comes first, and it is required.** Nothing is an agent until the law is met. Nothing downstream is defined without it.
2. **The kinematic laws of agency follow from the law.** They say how fast agency can move, per second and per joule, and what it must pay to move.
3. **Intelligence is the calculus of agency.** Intelligence is a hindsight characteristic. It is read off a completed run, the way velocity is read off a recorded path. It is therefore a derivative of agency: $\iota = dX/dJ$, where X is confirmed agency and J is joules.

In David's words: energy to run is the only true metric of computer intelligence. All other factors collapse to zero.

The order matters. A derivative needs a path. A path needs something that moves. The law says what moves. The kinematics say how it can move. Intelligence is read off the path afterward.

This track tests that order against the literature. It sorts every source into three bins.

- **Proven:** theorems and derivations. The result follows from stated assumptions.
- **Demonstrated:** experiments, measurements and simulations. The result was observed.
- **Opinion:** position papers, frameworks, definitions and philosophy. The result is argued.

No source is excluded for its method. Each source is placed by what it gives the law, the kinematics or the calculus.

Terms

- **SI:** the International System of Units.
- **Joule (J):** the SI unit of energy. One watt for one second.
- k_B : the Boltzmann constant, 1.380649×10^{-23} J/K, exact by SI definition.
- **Entropy production (Σ):** the total entropy a process creates, in units of k_B . It is zero only for a reversible process.
- **Mutual information (I):** the number of bits one variable carries about another.
- **Dynamical activity (A):** the rate of jumps between states in a stochastic process. It measures how busy the dynamics are.
- **TUR (thermodynamic uncertainty relation):** a theorem that ties the precision of a current to its entropy production.
- **NMR (nuclear magnetic resonance):** a method that controls and reads molecular spins.
- **Mixture of Limits:** the OpenIE philosophy. Use the cheapest gear that is sufficient.
- **Notational Intelligence:** the OpenIE receipt study. Every act carries its prediction, its outcome and its joules.
- **Metabolic Intelligence:** the OpenIE product. It runs on any fabric.
- **Klere:** the hardware.

2. Formal sketch

2.1 Layer 1. The law: eligibility

For a system S , an act a and an episode budget B in joules, eligibility $E(S, a) \in \{0, 1\}$ equals 1 only when all four clauses hold.

1. **Prediction first.** A result template $R^*(a)$ is fixed at time t_0 , before the act, and it discriminates: $D(R^*(a)) \geq \delta$.
2. **Coupling to the world.** $I(\text{do}(a); W') > 0$. Here W' is the world state downstream of the actuator, and $\text{do}(a)$ is Pearl's intervention operator [1]. The coupling must survive changes to the system's own inputs. Editing your own sensors does not count [2].
3. **Consequence borne.** The comparison $m = M(R, R^*(a)) \in \{1, \mu, 0\}$ (match, partial, mismatch) changes the system: $s_{t+1} = f(s_t, m)$.
4. **Physical floor.** $J(a) \leq B$. If no act meets clauses 1 to 4, the act is refusal, with a receipt.

Physics already prices clauses 1 to 3. The second law with feedback [3, 4] and the measurement and erasure bound [5] give, for one bath at temperature T :

$$W_{\text{ext}} \leq -\Delta F + k_B T I, \quad W_{\text{meas}} + W_{\text{erase}} \geq k_B T I$$

Here I is the mutual information the act's comparison acquires. An acceptor that compares and then resets pays at least $k_B T \ln 2$ per bit over a closed cycle [6, 7]. The law is physical from its first clause.

2.2 Layer 2. Kinematics

Cumulative joules are $J(t) = \int_0^t P(t') dt'$, with P the metered power.

Agency displacement counts bits of confirmed, coupled change:

$$X(t) = \sum_{k: t_k \leq t} E_k m_k I(\text{do}(a_k); W'_k)$$

Velocity per second is $v_t = dX/dt$. Velocity per joule is $v_J = dX/dJ$. Acceleration per joule is $\alpha_J = d^2 X/dJ^2$. The budget is $J(T) \leq B$.

The kinematic laws are the bounds on these quantities.

K1. Landauer floor, closed cycle. Each confirmed bit is recorded and later erased to reset the acceptor. So

$$v_J \leq \frac{1}{k_B T \ln 2}.$$

At $T = 300$ K, $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J, computed from the SI constants. That is about 3.5×10^{20} bits per joule.

K2. Classical speed limit. Shiraishi, Funo and Saito proved this bound for a Markov jump process with local detailed balance [8]:

$$\tau \geq \frac{L(p(0), p(\tau))^2}{2 \Sigma_{\text{tot}} \langle A \rangle_\tau}$$

- $L(p, p') = \sum_i |p_i - p'_i|$ is the full L_1 distance between two probability distributions. There is no factor of 1/2. With total variation distance $d_{TV} = L/2$, the same bound reads $\tau \geq 2d_{TV}^2 / (\Sigma_{\text{tot}} \langle A \rangle_\tau)$.
- $\Sigma_{\text{tot}} = \int_0^\tau \dot{\Sigma} dt$ is total entropy production, system plus baths, in units of k_B .
- $A(t) = \sum_{i \neq j} W_{ij}(t) p_j(t)$ is the dynamical activity, with W_{ij} the jump rate from state j to state i . $\langle A \rangle_\tau = \frac{1}{\tau} \int_0^\tau A(t) dt$ is its time average.

Rearranged, $\Sigma_{\text{tot}} \geq L^2 / (2\tau \langle A \rangle_\tau)$. Moving a distribution far, fast, with few jumps, costs entropy.

Joule form. For one bath, $k_B T \Sigma_{\text{tot}} = W - \Delta F_{\text{neq}}$, with F_{neq} the nonequilibrium free energy [9]. So the dissipated work obeys

$$W_{\text{diss}} \geq \frac{k_B T L^2}{2 \tau \langle A \rangle_\tau}.$$

Stationary currents. The same paper gives $\tau \geq c^* L^2 / (2 \Sigma_{HS} \langle A \rangle_\tau)$, with $c^* = 0.896 \dots$ and Σ_{HS} the Hatano-Sasa, or excess, entropy production [8].

K3. Precision costs dissipation. For a time-antisymmetric current Y in a steady state, $\text{Var}(Y)/\langle Y \rangle^2 \geq 2/\Sigma_{\text{tot}}$ [10, 11]. Agents use feedback, and feedback beats this bound in experiment [12]. The law for agents is the feedback form, in which mutual information enters the bound [13].

K4. Finite-time cost. Erasing a bit in finite time τ costs more than $k_B T \ln 2$. The excess grows as τ shrinks [14]. Near equilibrium, the excess work along a protocol is bounded below by the squared thermodynamic length over the duration [15].

K5. Quantum ceiling on rate. A system with mean energy $\langle E \rangle$ above its ground state makes at most $2\langle E \rangle/(\pi\hbar)$ orthogonal transitions per second [16]. That is about 6.04×10^{33} transitions per second per joule, computed from the constants. A figure of about 3×10^{33} matches $2E/h$, not $2E/(\pi\hbar)$.

2.3 Layer 3. Calculus: intelligence as a derivative, read in hindsight

A run produces a receipt: the ordered record $\{(J_k, X_k)\}_{k=0}^n$ of metered joules and confirmed displacement. Each m_k is known only after act k closes. So the path $X(J)$ exists only once the run is complete.

- **Intelligence, marginal:** $\iota(J) = \frac{dX}{dJ}$, estimated from the receipt as $\iota_k = \Delta X_k / \Delta J_k$.
- **Run integral:** $X(B) = \int_0^B \iota(J) dJ$.
- **Joules to close a requirement:** $J^*(X_{\text{req}}) = \int_0^{X_{\text{req}}} \frac{dX}{\iota}$.
- **Optimum:** $\pi^* = \arg \min_{\pi} J_{\pi}$ subject to $X_{\pi} \geq X_{\text{req}}$, $E_k = 1$ for every k , and $J_{\pi} \leq B$. If no policy is feasible, refuse with a receipt.
- **Ranking:** $S_1 \succ_q S_2 \iff J_{S_1}^*(q) < J_{S_2}^*(q)$.

X_{req} is set by the task's completeness predicate $C(z)$. It is a gate, not a score. Parameters, tokens and benchmarks enter only through J^* . All other factors collapse to zero.

Hindsight is literal here. Velocity is read off a recorded path. Intelligence is read off a recorded run. The receipt is the path.

2.4 Where the studies sit

- **Mixture of Limits** is the optimum operator. The cheapest sufficient gear is the argmin over joules.
- **Notational Intelligence** is the receipt. It carries R^* , R , m and J for every act.
- **Metabolic Intelligence** computes ι on any fabric.
- **Klere** is the hardware that meters J .

3. Layer 1. The Universal Law of Agency

The law has four clauses: prediction fixed before the act, coupling to the world through intervention, consequence borne by the system, and a physical floor in joules with refusal on a receipt.

3.1 Proven

Information has a price. Landauer showed that a logically irreversible operation dissipates at least $k_B T \ln 2$ per bit [6]. This is the floor under clause 4. Bennett built a reversible Turing machine and showed that only erasure must cost [17]. The unavoidable cost of agency sits at the reset of the acceptor, not at the thinking. Bennett then resolved Maxwell's demon by the cost of erasing the demon's memory [7]. The demon is a minimal agent. Its accounts close only when its memory is reset. That is clause 3 plus clause 4.

Feedback is priced by information. Sagawa and Ueda proved that with feedback, extractable work is bounded by $-\Delta F + k_B T I$ [3]. The value of acting on a measurement is capped by the information the measurement carries. They proved that measurement work plus erasure work is at least $k_B T I$ [5]. The full cycle of predict, compare and reset is priced in joules. They derived an exact Jarzynski equality under feedback [4] and a fluctuation theorem with information exchange between subsystems [18]. Coupling is a correlation that enters the second law.

Entropy converts to joules. Esposito and Van den Broeck derived $k_B T \Sigma = W - \Delta F_{\text{neq}}$ for an isothermal driven system [9]. Every kinematic bound in Layer 2 becomes a joule bound through this identity. Horodecki and Oppenheim proved work costs for single shots at small scale [19]. Agency acts are single shots. The standard reviews state the settled account: information is a thermodynamic resource with a price [20, 21].

Control needs coupling. Conant and Ashby proved that a good regulator must be a model of the system it regulates [22]. A system that regulates well already carries a prediction. Touchette and Lloyd proved that the entropy a controller removes is bounded by its mutual information with the system [23, 24]. Coupling is the currency of control. Closed loop control, acting on a comparison, is what the law requires. Pearl's do-calculus gives the intervention operator and the conditions under which its effects are identifiable [1]. Clause 2 is written in this operator. Everitt and colleagues classified how an agent can tamper with its own reward or inputs [2]. Coupling must be to the world, not to the agent's own sensor.

Information flows inside the agent. Horowitz and Esposito showed that information flow between subsystems enters each subsystem's second law [25]. Coupling becomes a rate. Ito and Sagawa proved a second law for arbitrary causal networks with transfer entropy [26]. Prediction, act and outcome are nodes on such a network. Hartich, Barato and Seifert linked transfer entropy, learning rate and dissipation [27]. Allahverdyan, Janzing and Mahler bounded the efficiency of information flow between coupled systems [28]. Esposito and Schaller showed that feedback that changes only barriers still enters the entropy balance [29]. A purely informational act is in the joule ledger.

Autonomous demons. Mandal and Jarzynski solved a demon that lifts a mass by writing bits to a tape [30]. It has no outside operator, and memory pays for its work. Boyd, Mandal and Crutchfield bounded the work of such ratchets by the entropy rates of their input and output tapes [31]. They showed that modular designs pay dissipation above Landauer [32]. How an agent is built changes its floor.

Prediction quality has a joule price. Kolchinsky and Wolpert proved that a process tuned for one input distribution dissipates extra on any other, by a relative entropy term [33]. Still, Sivak, Bell and Crooks showed that memory that does not predict costs dissipation [34]. Keeping only what predicts is the cheap design. Kolchinsky and Wolpert defined semantic information as information that is causally necessary for a system to keep itself out of equilibrium, with a thermodynamic value [35]. It is the closest existing law with joules. It stops before kinematics and before an intelligence operator. Ortega and Braun derived bounded-rational decisions from a free-energy principle with an information cost [36]. Kolchinsky and Wolpert bounded the heat of universal computation by program complexity [37].

3.2 Demonstrated

Landauer's floor, measured. A colloidal bead in an optical double well approached $k_B T \ln 2$ of mean heat as erasure slowed [38]. A feedback trap measured the same approach with high precision and extra cost at short cycle times [39]. The floor rises when the act is fast. That is a Layer 2 effect seen in a Layer 1 experiment. An adiabatic electronic circuit showed the Landauer cost for irreversible operation and dissipation below $k_B T$ for reversible operation [40]. The floor binds the reset, not the logic. Nanomagnetic memory bits showed the floor in memory hardware [41]. Erasure with near-zero work appears when the free-energy accounting is complete [42]. The floor is on total entropy, so the receipt must carry the full account. The entropy that enters the bound was measured to have the Gibbs-Shannon form [43]. A single trapped ion showed the quantum floor [44]. The floor holds on every substrate tested. An underdamped micromechanical oscillator reached the floor at high speed [45]. A quantum dot erased bits at minimal dissipation for a given duration, with protocols designed from thermodynamic length [46].

Feedback agents turn information into work. A colloid on a spiral staircase potential gained free energy above the work done, paid by information, and confirmed the generalized Jarzynski equality [47]. This is the first direct observation of an act whose value is set by a measurement made before it. A single-electron box extracted work near $k_B T \ln 2$ per bit [48] and measured the fluctuation theorem with mutual information [49]. An autonomous on-chip demon cooled a system with information while its own heating was measured [50]. The agent's cost was measured beside its effect.

Quantum agents obey the same law. NMR on molecular spins reduced entropy production by feedback, within the information bound [51], and converted information to energy at the floor [52]. Superconducting circuits tracked a demon's memory during feedback [53], verified the

generalized integral fluctuation theorem [54], and measured information gained and lost during feedback [55]. Coupling decays, so the receipt must record it at the time of the act. Cold atoms in a three-dimensional optical lattice were sorted by measurement [56].

Classical engines measured the price at full scale. An optical tweezer engine converted information to work at the theoretical limit [57]. A colloidal engine extracted work that depends on temporal correlations in its input [58]. A DNA hairpin under continuous monitoring yielded more than $k_B T \ln 2$ per cycle when many measurements happened per cycle, consistent with the information content [59]. A levitated microparticle with delayed feedback obeyed a generalized second law across two decades of delay, and feedback failed at large delay [60]. A late act is a weaker act. Feedback trap engines maximized power and velocity [61], exploited noisy measurements optimally [62], and harvested a nonequilibrium bath [63]. A quantum dot linked work fluctuations to dissipation during information-to-work conversion [64]. A trapped ion verified fluctuation theorems that include the demon's own dissipation [65], following theory by Zeng and Wang [66].

Functional-system agents built and run. A simulated nematode with an acceptor of action results learned locomotion and chemotaxis [67]. A physical mobile robot ran the same functional-system controller [68]. Clauses 1 and 3 run in software and on a body. Neither meters joules.

3.3 Opinion

Purposive systems and anticipation. Rosenblueth, Wiener and Bigelow defined purposeful behavior through negative feedback toward a goal [69]. It has no energy term. Anokhin's functional system has afferent synthesis, decision, an acceptor of action results fixed before the act, action, and comparison [70, 71]. This is clauses 1 and 3 in their first systematic form, built on animal experiments. Sudakov stated the result as the system-forming factor [72]. The prediction clause is the organizing principle. Shvyrkov showed from single-neuron recordings that the goal organizes neural activity [73]. Rosen defined an anticipatory system as one that contains a predictive model of itself and its world and acts on the prediction [74]. None of these sets a joule floor.

Autonomy and organization. Maturana and Varela defined the living by self-production [75]. Kauffman put a thermodynamic work cycle inside the definition of an autonomous agent [76]. It has no prediction clause. Di Paolo made adaptivity, the regulation of one's own viability, a condition for sense-making [77]. That is clause 3. Barandiaran and Moreno proposed a minimal criterion for cognitive organization and grounded adaptive regulation in metabolism [78, 79]. Barandiaran, Di Paolo and Rohde defined agency by individuality, interactional asymmetry and normativity [80]. Ruiz-Mirazo, Moreno and Mossio framed autonomy as a self-maintaining metabolic organization [81, 82]. Montévil and Mossio described organisms as closures of constraints on thermodynamic flows [83]. It is the closest of this school to joules. Aguilera and Barandiaran built a minimal model of autonomous agency in stochastic thermodynamics [84]. The autonomy tradition is moving toward the joule.

Free energy and the physics of agency. Friston proposes that agents minimize variational free energy [85, 86]. That free energy is informational, not joules, and every bounded system qualifies, so it sets no floor. Rovelli explains agency through the entropy gradient [87]. Agency is thermodynamic at root. Jaeger argues that agency requires self-manufacture and that current artificial intelligence lacks it [88]. That is a demand for a gate before any intelligence claim. Azadi ties agency to computational irreducibility [89]. Wissner-Gross and Freer propose a force toward future path diversity [90]. Its causal temperature is not a bath temperature, so its units are not joules. Kappen's comment critiques it [91].

4. Layer 2. The kinematic laws of agency

Once an act is eligible, physics bounds how fast agency moves per second and per joule, and what precision and speed cost. The bounds come in four families: speed limits, the TUR, finite-time costs and rate ceilings.

4.1 Proven

Speed limits. Shiraishi, Funo and Saito proved the classical speed limit in Section 2.2 [8]. Moving the agent's state distribution a distance L in time τ costs entropy, and so joules. The bound extends to open quantum systems [92] and to strong coupling with general environments [93]. Vo, Van Vu and Hasegawa derived the speed limit and the TUR from one inequality [94]. Falasco and Esposito bounded the time to complete a transition by the dissipation spent [95]. Ito bounded the information-geometric speed of a distribution by entropy production [96]. Ito and Dechant bounded the rate of change of any observable by Fisher information and dissipation [97]. Nakazato and Ito bounded entropy production by the Wasserstein distance traveled [98]. Dechant, Sasa and Ito split entropy production into the part that moves the system and the part that keeps it running [99]. Van Vu and Saito unified the TUR, minimum dissipation and speed limits through optimal transport [100]. Nagayama, Yoshimura and Ito derived a family of speed limits indexed by generalized means of activity [101]. Yoshimura and Ito proved the TUR and the speed limit in deterministic chemical reaction networks [102]. Aurell, Mejía-Monasterio and Muratore-Ginanneschi solved minimal finite-time dissipation as an optimal transport problem [103]. The kinematic laws form one family. The cheapest way to move an agent's state is a transport map.

Precision costs dissipation. Barato and Seifert stated the TUR, $\text{Var}(Y)/\langle Y \rangle^2 \geq 2/\Sigma$ [10]. A reliable agent pays for its reliability. From $2/\epsilon^2$ at $\epsilon = 0.01$, precision of 1% costs at least $20,000 k_B T$ [104]. Gingrich, Horowitz, Perunov and England proved the TUR for all steady-state currents [11]. Horowitz and Gingrich proved it for finite observation times [105], which is what agents have. Further results bound the full distribution of fluctuations [106], cover discrete time steps [107], extend the TUR wherever a fluctuation theorem holds [108], and cover Langevin dynamics, underdamped motion and velocity feedback [109, 110, 111]. Correlations between currents tighten the bounds [112]. Potts and Samuelsson showed that any fluctuation relation implies a TUR, including with measurement and feedback [13]. This is the precision law for agents. A

review states the field [113]. Barato and Seifert priced the precision of a clock [114]. Salazar bounded entropy production by information from the detailed fluctuation theorem [115]. Landi and Paternostro reviewed entropy production from classical to quantum [116].

Finite-time costs. Schmiedl and Seifert found optimal finite-time protocols, with jumps at the ends [117]. The optimal act at finite speed is not the slow act. Sivak and Crooks bounded excess work by the squared thermodynamic length over the duration [15]. Proesmans, Ehrich and Bechhoefer found the minimal cost of erasing a bit in time τ [14, 118]. The Layer 1 floor becomes a Layer 2 curve. Zhen and colleagues stated a universal finite-time bound on reset energy [119]. Speed and efficiency trade against each other by theorem: efficiency at maximum power [120, 121], finite power forbidding Carnot efficiency [122], a three-way trade among power, efficiency and constancy [123], and a universal constraint for low-dissipation engines [124]. Tsirlin and colleagues found minimal dissipation at a fixed rate [125].

Rate ceilings. Margolus and Levitin bounded orthogonal transitions per second by energy [16]. Lloyd applied the limit to a whole computer [126]. Deffner and Campbell reviewed the family of quantum speed limits [127].

Living sensors. Energy dissipation, adaptation speed and accuracy are bound together in bacterial chemotaxis [128]. Learning about the environment requires energy [129], and inference accuracy is tied to dissipation [130]. Sensing precision is limited by receptors, time and energy, with an optimal allocation among them [131]. No resource can be skipped. That fits Mixture of Limits. Erasure inside adaptation has a cost [132]. Information acquired per energy dissipated is the nearest existing quantity to v_J in living sensors [133, 134]. Copying information costs energy [135]. Writing the receipt has a price. Accurate and synchronized biochemical clocks cost free energy [136, 137], and so does reducing noise at high sensitivity [138]. Holding a system away from equilibrium has a minimum power [139]. That is the standing cost of an agent ready to act.

Information rates without joules. Schreiber defined transfer entropy, directed information flow per step [140]. It is per time, not per joule, and it has no eligibility gate. Tishby and Polani charged information per step in a Bellman recursion [141]. Landauer converts those bits to joules. Stratonovich bounded the maximal gain from a given amount of information [142]. That is a ceiling on value per coupled bit.

4.2 Demonstrated

The TUR tested. A two-qubit NMR experiment obeyed generalized TURs and violated the specialized TUR where theory predicts [143]. An optical tweezer information engine violated the original TUR near maximal efficiency and obeyed generalized bounds with mutual information [12]. Agents need the feedback form. A quantum dot Szilard engine confirmed it independently [144]. Atomic-scale conductors examined the TUR with measured current noise, the physics of every chip [145]. Molecular motors were scored by precision per unit dissipation [146]. Observed current fluctuations bound dissipation from below [147]. A receipt of fluctuations is a lower-bound meter for joules.

Speed limits and finite-time costs tested. Single atoms in an optical lattice showed the Mandelstam-Tamm and Margolus-Levitin bounds and the crossover between them [148]. The rate ceiling is measured. A trapped colloid measured time and entropy trade-offs for fast thermal transitions [149]. Brownian particles driven by optimal transport protocols saturated the minimal finite-time dissipation bound [150]. The kinematic optimum is reachable. A feedback trap measured a two-force Brownian machine against linear response bounds [151]. A photonic experiment tested geometric bounds on entropy production [152].

Heat engines at finite speed. A single colloid ran a micrometer Stirling engine [153]. A trapped bead ran a Brownian Carnot cycle with efficiency at maximum power measured [154]. The energy cost of choosing one of two options was measured [155]. That is the smallest act. A colloidal engine ran on an active bacterial bath [156], and engineered noise gave near-Carnot efficiency at finite power by shortening relaxation [157]. Engineering activity moves the bound. An information engine flipped from refrigerator to heater with noise [158]. The receipt must record the regime, not only means.

Empowerment in simulation. Klyubin, Polani and Nehaniv defined empowerment, the channel capacity from actions to future sensors [159]. It was computed for continuous control [160] and reviewed [161]. It measures how much an agent can move the world, in bits. It is not a rate per joule.

4.3 Opinion

Kinematics is mostly theorem and experiment. The opinions here concern which variable is right. Seifert holds that inference from fluctuations is the route to hidden costs [104]. Rovelli and Wissner-Gross and Freer both propose entropic kinematics [87, 90]. Kolchinsky shows that dissipation alone does not bound how fast replicators grow or decay [162]. Rates are bounded by dissipation together with activity. That is the form of the speed limit, where $\langle A \rangle$ sits beside Σ .

5. Layer 3. Intelligence as the calculus of agency

Intelligence is the derivative of gated agency in joules, read off a completed run. This layer needs three things from the literature: theory that ties learning and decision to joules, measurement of joules per unit of useful work, and positions on what intelligence is.

5.1 Proven

Learning and prediction priced in joules. Goldt and Seifert bounded the information a learning network acquires by the entropy it produces [163]. Learning efficiency is at most one. This is ι for the learning part of an act, with a ceiling. Still showed that optimal memory keeps only predictive information [164]. The marginal joule spent on memory must buy prediction. Boyd, Crutchfield and Gu proved that the agent that extracts the most work from data has the maximum-likelihood model of that data [165]. Better models yield more work per joule. Boyd, Crutchfield, Gu and Binder stated overfitting and generalization as energetics [166]. Generalization appears as joules. Ehrich, Still and Sivak priced the controller itself, beyond the information it uses [167]. So J must

be metered at the wall of the whole system. Ehrich and Sivak gave the ledger of energy and information flows in autonomous machines [168]. Wolpert reviewed computation costs beyond Landauer [169]. Levy and Baxter showed that neural codes that maximize bits per unit energy differ from codes that maximize bits [170]. The right objective for a brain is information per joule.

Decisions and bounded optimality. Russell and Subramanian defined bounded optimality: the best program for a given machine and environment [171]. Here the resource is joules, and the optimum is $\arg \min J$ subject to $X \geq X_{\text{req}}$. Genewein and colleagues derived abstraction and hierarchy from utility minus information cost [172]. Mixture of Limits gears are such a hierarchy. Legg and Hutter defined intelligence as complexity-weighted expected reward over all computable environments [173]. It is an integral, but over reward, without energy, and uncomputable. The derivative in joules replaces reward with confirmed coupled bits and the measure with metered joules. Takahashi and Hayashi define empowerment per joule from stochastic thermodynamics [174]. It is the closest existing calculus in joules. It has no law gate in front of it.

5.2 Demonstrated

Brains. Laughlin, de Ruyter van Steveninck and Anderson measured the energy cost per bit in blowfly photoreceptors and interneurons [175]. Higher information rates cost more per bit. That is a measured ι in a living agent and a measured diminishing return, $\alpha_J < 0$ in sensing. Attwell and Laughlin built the energy budget of grey matter by process [176]. Lennie showed that energy limits how many cortical neurons can be active at once [177]. Energy is the constraint, and the rest adapts. Energy per action potential differs widely across neuron types [178]. Same function, different joules. Neural design follows energy efficiency [179].

Machines. Memory access costs far more energy than arithmetic [180]. Parameter count matters only through the joules it makes you move. Intelligence per watt measures task accuracy per unit power on local accelerators [181]. Its numerator is benchmark accuracy, not gated agency. MLPerf Power standardizes power measurement from microwatts to megawatts [182]. ML.ENERGY measures inference energy automatically [183]. TokenPowerBench benchmarks the power of large language model (LLM) inference [184]. Jin, Wei and Brooks analyze the energy of test-time compute [185]. General-purpose models cost much more energy per task than task-specific ones [186]. The cheapest sufficient model wins in joules. That is Mixture of Limits, measured.

Meters. RAPL (Running Average Power Limit) is the set of on-die energy counters in Intel and AMD processors. Its readings were validated against external measurement [187]. DRAM (dynamic random-access memory) readings were validated separately [188]. Meter quality depends on the processor generation [189]. Apple's powermetrics manual states that its average power values "are estimated and may be inaccurate" and should not be used to compare devices [190]. On Apple silicon, powermetrics readings are reported values. Measured joules for that machine come from an external wall meter.

Compression scored. The Hutter Prize scores lossless compression of the first gigabyte of an English Wikipedia dump, with a prize fund of 500,000 euros and limits on runtime and memory [191]. It bounds time and memory, never joules. A joule cap would make it a calculus-of-agency benchmark for one task class.

Estimates. These sources are valuable, and their joule figures are modeled, extrapolated or computed. They sit beside measured joules, never in their place. Estimates are never measured_j. Strubell, Ganesh and McCallum sampled power for short runs and extrapolated to full training of natural language processing models [192]. Patterson and colleagues computed training energy from reported hardware, runtime and data-center data [193]. Lacoste and colleagues built a calculator from hardware type, runtime and region [194]. The BLOOM footprint mixes metered and modeled inputs [195]. Carbontracker reads on-device counters and predicts full-run totals [196]. Its predictions are estimates.

5.3 Opinion

Each position is placed against the definition: intelligence is the derivative of gated agency in joules, read off a completed run.

- Schmidhuber treats intrinsic reward as compression progress, a first derivative of how well an agent compresses its history [197]. It is the closest precedent for intelligence as a derivative read from a run. Its variable is compression, not joules.
- Chollet defines intelligence as skill-acquisition efficiency over priors and experience, with the ARC (Abstraction and Reasoning Corpus) benchmark [198]. It is a ratio. Its denominator is information and data, not joules.
- Hernández-Orallo builds universal psychometrics across species and machines [199]. It evaluates by tasks. This track evaluates by joules to close tasks.
- Gershman, Horvitz and Tenenbaum define intelligence as expected utility net of computation cost [200]. This track makes the cost joules.
- Lieder and Griffiths treat cognition as optimal use of limited resources [201]. It is close to Mixture of Limits. The resource is left abstract.
- Balasubramanian reads the brain's design through its energy limits [202].
- Silver, Singh, Precup and Sutton hold that reward is enough [203]. Reward is chosen by a designer. Joules are read off a meter. Vamplew and colleagues reply that scalar reward cannot express multi-objective goals [204]. This track keeps one variable, joules, and moves multiplicity into the gate $X \geq X_{\text{req}}$.
- Hafez and colleagues define agency and intelligence together [205]. It is not joule-native.
- Karagoz makes energy self-sustainment the objective [206]. It has no prediction gate, no kinematics and no joule ranking across systems.

5.4 Where the calculus stands

- **Proven:** learning, memory, prediction and control each have a joule price with a ceiling.
- **Demonstrated:** joules per bit are measured in brains. Joules per task are measured in machines. Meters can be validated.

- **Missing everywhere:** no source reads intelligence as dX/dJ off a gated, completed run. That is David's contribution and the work of this track.

6. Global findings, by method

The search ran in Russian, Chinese, Japanese, German, French, Spanish, Portuguese, Korean, Italian, Polish, Hindi and Arabic, with English follow-ups. Results are organized by method and class of result. Language is metadata only. No entry is ranked by where it came from.

6.1 Theorems and reviews: information has a thermodynamic price

- Poplavskii gave an early systematic treatment of the energy cost of acquiring and processing information [207]. Written in Russian, with an English translation. Layer: law.
- Stratonovich proved value-of-information theorems [142]. English edition of a Russian monograph. Layer: kinematics.
- Ito and Sagawa applied information thermodynamics on networks to E. coli chemotaxis. Transfer entropy bounds robustness, and information-thermodynamic efficiency is high where ordinary thermodynamic efficiency is low [208]. Written in Japanese. Layers: law and kinematics.
- Sun and Quan reviewed Maxwell's demon and the physical floor on dissipation in information processing [209]. Written in Chinese. Layer: law.
- Quan, Dong and Sun reviewed mesoscopic thermodynamics: the demon is consistent with the second law once erasure is counted, and power-efficiency constraints were tested [210]. Written in Chinese with an English abstract. Layers: law and kinematics.
- Parrondo's encyclopedic review gives the current account of information as a thermodynamic resource [211]. Layer: law.
- Strasberg's doctoral thesis tests common assumptions in information thermodynamics with physical models [212]. Written in English with a German abstract. Layer: law.
- Further theorems surfaced through searches in Japanese, Russian, Chinese, Korean, Portuguese, Italian and Polish: speed limits and TURs in chemical networks [102, 97, 101], finite-time thermodynamics [125, 124, 121], Langevin TURs [110, 111], information bounds on entropy production [115, 116], and single-shot work costs [19].

6.2 Experiments: the price measured

- Dago's doctoral thesis reports 1-bit erasure and writing near the minimal energy at high speed on an underdamped micro-cantilever [213]. Written in French. Layer: the law's floor reached at kinematic speed.
- Lagoin's doctoral thesis builds macroscopic Maxwell's demons, including a Szilard engine, from a vane in a granular gas [214]. Written in French. Layer: law. The demon's accounting holds outside the microscopic regime.
- Ciampini and colleagues used extractable work to witness quantum correlations in a photonic experiment [215]. Layer: law.

- Experiments surfaced through searches in Korean, Portuguese, Hindi, Italian, Spanish, Chinese and Japanese: information engines [57, 12], NMR demons [51, 52], the TUR on qubits [143], active-bath engines [156, 157], geometric bounds [152], symmetry breaking and Carnot cycles [155, 154], trapped-ion demons [44, 65], and feedback conversion and optimal transport [47, 54, 150].

6.3 Theory of functional systems: prediction before the act

The Anokhin school states clauses 1 and 3 in physiological terms. A result template, the acceptor of action results, is formed before the act, compared after it, and the comparison reorganizes behavior. These works were found through Russian-language searches.

- Anokhin's functional system [70, 71].
- Sudakov on the result as the system-forming factor and on probabilistic prediction of behavior [72, 216, 217, 218].
- Shvyrkov on goals organizing neuronal activity and on learning as selection of neuronal systems [73, 219].
- K. V. Anokhin on the brain's cognitive structure as a hypernetwork of functional-system elements [220].
- Saltykov and Grachev on the system-forming factor as anticipation [221].
- Vityaev on purposefulness formalized as rule learning that predicts results [222].

Implemented and run. Cognitive architectures built on functional-system theory form result templates, compare, and learn subgoals, in simulation and on a physical robot [223, 67, 68, 224, 225]. None meters joules. Adding a meter to one of these agents would test the full order of law, kinematics and calculus on an existing implementation. Differentiable probabilistic logic networks are related calculus tooling, not built on functional systems, with no law gate and no joules [226].

6.4 Autonomy, organization and minimal agency

Found through Spanish, French and German searches. Agency is a self-maintaining organization that regulates its interactions [82, 81, 78, 79, 80, 77]. The organization is a closure of constraints on thermodynamic flows [83]. The autonomy school has stepped into stochastic thermodynamics [84]. Layer: law, as the organization that must exist before any act counts.

6.5 Measured joules for machine intelligence

Found through Chinese-language searches on inference energy: automated inference energy measurement [183], per-token power benchmarks [184], and the energy of test-time reasoning [185].

6.6 Philosophy of agency

Rovelli's physics of agency surfaced through German [87]. Jaeger and Azadi complete the set of position papers [88, 89].

6.7 Searches with no primary research

Hindi and Arabic searches returned no primary research in those languages. English follow-ups surfaced work from Indian institutions, listed above by method.

7. Open problems

7.1 What must be proven

P1. The run path is well defined. For any run with metered power $P(t) > 0$, X is a function of cumulative joules J , and $\iota = dX/dJ$ exists almost everywhere. In hand: $E_k \in \{0, 1\}$, $m_k \in \{0, \mu, 1\}$ and $I \geq 0$, so X never decreases. A monotone function is differentiable almost everywhere by Lebesgue's theorem. $P > 0$ makes $J(t)$ strictly increasing. Missing: convergence of $\iota_k = \Delta X_k / \Delta J_k$ to ι as meter sampling refines, with an error term set by meter resolution and sampling interval.

P2. Agency-gated Landauer bound. Under closed-cycle accounting, each confirmed coupled bit costs at least $k_B T \ln 2$, so $v_J \leq 1/(k_B T \ln 2)$. In hand: measurement plus erasure work is at least $k_B T I$ [5], and erasure costs at least $k_B T \ln 2$ per bit [6, 7]. Missing: a data-processing step showing that the coupled bits credited in X cannot exceed the bits recorded in the acceptor's memory during the comparison. Clause 3 already requires the confirmation m to come from a physical record.

P3. Agency speed limit. The time and dissipated joules to close act k are bounded below by a function of the coupled bits it confirms. In hand: the speed limit and its joule form [8], and Pinsker's inequality, $D_{\text{KL}}(p||q) \geq \frac{1}{2} \|p - q\|_1^2$ in nats, where D_{KL} is the Kullback-Leibler divergence. Missing: a lower bound on the L_1 displacement of the joint agent-world distribution in terms of $I(\text{do}(a); W')$. That needs a reverse inequality under stated conditions, such as a floor on the smallest state probability.

P4. Precision law for agents. $\text{Var}(X)/\langle X \rangle^2$ over a run is bounded below by a function of entropy production and the information used in feedback. In hand: the standard TUR fails for feedback agents in experiment [12], and the feedback form holds [13]. Missing: whether X is a time-antisymmetric current, or which current it bounds, and the feedback TUR written in the receipt's variables.

P5. Hindsight theorem. ι is a function of the completed receipt only. No pre-run quantity fixes it. In hand: each m_k compares an outcome to a prediction fixed before the act, so m_k is unknown before act k closes. Missing: a construction of two worlds that agree on everything a pre-run predictor sees and differ in some m_k . With P1, this makes "intelligence is read off the completed run" a theorem.

P6. Ranking survives meter error. If $J_{S_1}^*(q) < J_{S_2}^*(q) (1 - \epsilon)/(1 + \epsilon)$, with ϵ the meter's relative error bound, then $S_1 \succ_q S_2$ under any reading within that error. In hand: direct from the definition. Missing: validated ϵ for each meter [187].

P7. Mixture of Limits optimality. The cheapest-sufficient rule tries gears from cheapest up, stops at the first gear whose output passes $C(z)$, and refuses with a receipt if the next gear would break the budget. It achieves the joule minimum up to the cost of failed attempts. In hand: if sufficiency is monotone in gear order and gear costs rise with order, the first sufficient gear is the cheapest sufficient gear. Missing: a regret bound when sufficiency is not monotone, and the conditions under which this rule and the σ -law selector, $a^* = \arg \max[H(a) - \lambda J(a)]$ subject to $J \leq B$, choose the same act.

P8. When agency compounds. Conditions under which $\alpha_J > 0$ over a run, so that earlier receipts lower the joules of later closures. In hand: an earlier OpenIE toy run found $\alpha_J \leq 0$ on near-optimal tasks. Missing: a model of reuse in which stored receipts cut later J by more than the storage and lookup joules. Still gives the memory cost side [164].

7.2 What must be demonstrated

ID	Claim	Evidence required
D1	J^* per task is measured, not estimated, on at least two fabrics	measured_j from RAPL on x86 Linux and an external wall meter on Apple silicon
D2	The σ -law selector closes tasks in fewer measured joules than the Mixture of Limits rule under binding budgets	Paired runs with the same tasks, gears, predicates, budgets and meters
D3	Refuse with a receipt works	Zero overspends, refusals on infeasible tasks, few refusals on feasible ones
D4	ι is read off receipts and is stable across seeds	Per-run ι_k series with spread within a pre-set bound
D5	Estimates do not substitute for measurement	Rank agreement between est_j and measured_j, reported, never merged
D6	The physical floor is respected and the gap is known	Measured joules per confirmed bit divided by $k_B T \ln 2$

8. Experiment plan

Every table, receipt and figure in this plan follows one rule. Estimates are never measured_j. A modeled figure is labeled est_j. A metered figure is labeled measured_j and names its meter.

8.1 Meters

Acronyms: CPU (central processing unit), GPU (graphics processing unit), ANE (Apple Neural Engine), AC (alternating current), NVML (NVIDIA Management Library).

Tier	Meter	What it reads	Label
M0	External wall-plug power meter, logging at 1 Hz or faster	Whole-system AC power	measured_j, wall
M1	RAPL through Linux powercap, package, core, uncore and DRAM domains where present	Cumulative on-die energy in microjoules	measured_j, rapl
M1g	NVML total energy counter on a discrete GPU	Cumulative GPU board energy	measured_j, nvml
M2	Apple powermetrics, CPU and GPU samplers at 100 ms	CPU, GPU and ANE power as reported by the system	reported_j, powermetrics

- **RAPL.** Read the counter before and after each act. Handle wraparound. Validate against M0 on a calibration workload [187, 188].
- **powermetrics.** Readings are filed as reported_j. Measured joules on Apple silicon come from M0. Within one machine, reported_j attributes joules to CPU, GPU and ANE.
- **Idle baseline.** Record 120 s of idle power on every meter before each block. Report gross and idle-subtracted joules side by side. Never report only the subtracted figure.
- **State control.** Fix the CPU frequency governor where the operating system allows. Log ambient temperature and thermal pressure. Close background jobs. Discard one warm-up run per arm.

8.2 Tasks

Each task has a machine-checkable completeness predicate $C(z)$: unit tests for code, exact match for arithmetic, schema validation for extraction, a verified final state for tool use. Two strata are fixed in advance. Near-optimal tasks are those the cheapest gear usually closes. Compounding tasks are sequences where earlier receipts can be reused. X_{req} is set by $C(z)$. Passing is a gate, not a score.

8.3 Arms

All arms share the gears, the predicate, the budget, the meters and a randomized interleaved run order.

1. **σ -law selector:** $a^* = \arg \max [H(a) - \lambda J(a)]$ subject to $J \leq B$, with refusal on a receipt.
2. **Mixture of Limits cheapest-sufficient rule:** the fair rival. Same information, gears, budget and refusal right. Only the selection rule differs.
3. **Largest gear always:** reference ceiling.
4. **Smallest gear always:** reference floor, with refusal when it fails.

8.4 Budgets, seeds and statistics

- A pilot runs arm 2 without a budget and records the median measured joules per task, $\tilde{J}_q$.

- Budget levels are $B \in \{0.25, 0.5, 0.75, 1.0\} \times \tilde{J}_q$, plus unconstrained. A level binds only if at least 20% of tasks exceed it in the pilot. Levels that do not bind are reported and excluded from the D2 test.
- At least 10 seeds per arm, task and budget. More if the pilot shows 10 cannot resolve a 10% difference in median J^* at 95% confidence.
- Report mean, median, standard deviation and a 95% bootstrap confidence interval of J^* per task. Compare arms 1 and 2 with a Wilcoxon signed-rank test on per-task median J^* , with effect size.

8.5 Receipt schema

The receipt is the Notational Intelligence record of the run. One record per act:

```
act_id, run_id, task_id, arm, seed, budget_j, gear
prediction_hash, prediction_ts      # R*, fixed before the act
outcome_hash, outcome_ts           # R
m                                   # 1, mu, or 0
coupled_bits_est                    # estimate of I(do(a); W')
t_start, t_end
measured_j    {meter, domains, gross_j, idle_w, idle_subtracted_j}
reported_j    {meter, cpu_j, gpu_j, ane_j}      # powermetrics only
wall_j        {meter, gross_j, idle_w, idle_subtracted_j}
est_j         {model, inputs}                  # never merged with measured_j
cumulative_measured_j
refused, refuse_reason
passed_C
```

8.6 Metrics

- J^* per task: measured joules to pass $C(z)$, from M0 where present, otherwise M1.
- Closure rate per arm and budget.
- Overspends: acts that exceeded B . The target is zero.
- Refusal quality: precision and recall of refusals against tasks shown infeasible within B by any arm.
- Intelligence series: $\nu_k = \Delta X_k / \Delta J_k$ per run, and its spread across seeds.
- Acceleration sign: the sign of α_J from regressing ν_k on cumulative J , per stratum.
- Estimate check: Kendall's rank correlation between `est_j` and `measured_j`.
- Distance to floor: measured joules per confirmed coupled bit divided by $k_B T \ln 2$ at the logged temperature.

8.7 Falsifiers, fixed before any run

ID	Observation	Claim defeated
F1	The 95% interval for the paired difference in median J^* , σ -law minus Mixture of Limits, includes or exceeds zero at every binding budget	The σ -law selector beats its fair rival in joules
F2	Any arm records an overspend	That arm's refusal implementation
F3	Refusal recall on infeasible tasks is below 0.9, or more than 10% of refusals fall on tasks another arm closed within B	Refusal correctness
F4	Kendall's correlation between <code>est_j</code> and <code>measured_j</code> is below 0.8	Use of that estimator for any ranking
F5	The coefficient of variation of mean ι across seeds exceeds 0.25 on a task	Reading ι at that task's granularity
F6	The fitted α_J is not positive with 95% confidence on the compounding stratum	Compounding on those tasks
F7	Any act shows measured joules per confirmed bit below $k_B T \ln 2$	The accounting itself: a meter or book-keeping fault, fixed before any other result is reported
F8	M1 and M0 disagree on a calibration workload by more than the error bound in P6	That RAPL domain as a measured source on that machine

8.8 Reporting

Publish all receipts, raw meter traces, idle baselines and analysis code with the result. Report measured, reported and estimated joules in separate columns. Never sum across tiers. State which falsifiers fired, with the same prominence as those that passed.

8.9 Where the products sit

- **Klere** is the hardware that meters J . M0 and M1 stand in for it until Klere meters are on the bench.
- **Metabolic Intelligence** runs the arms on any fabric and computes ι from receipts.
- **Notational Intelligence** is the receipt format in Section 8.5.
- **Mixture of Limits** is the philosophy and the rival rule in arm 2.

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Entries marked "preprint" are cited by their arXiv posting. Entries published in a language other than English say so.

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